

DMS613: Introduction to Mathematical Finance

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Black-Scholes-Merton Formula

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1 Financial Markets

1.1 What is a Financial Market?

A financial market is a system in which participants trade financial assets. These participants include households, firms, banks, institutional investors, governments, and professional traders. Financial markets are broadly classified into:

- **Primary markets:** where new securities are issued (for example, an initial public offering).
- **Secondary markets:** where existing securities are traded among investors.

Our mathematical modeling concerns secondary markets, where prices evolve continuously in time and are observed dynamically.

1.2 Asset Classes

Financial assets can be grouped into several broad categories.

1.2.1 Equities (Stocks)

A stock represents ownership in a company. If an investor holds one share of a firm, they own a fraction of that firm's future profits and possibly receive dividends.

The stock price fluctuates randomly over time. In continuous-time modeling, we will assume that the stock price $(S(t))_{t \geq 0}$ follows a stochastic differential equation of the form

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t),$$

where μ represents the expected rate of return and σ represents volatility.

1.2.2 Fixed Income (Bonds)

A bond is a contract under which the issuer promises to make specified future payments.

The simplest fixed-income asset in continuous-time modeling is the *money market account*, which evolves deterministically:

$$B(t) = e^{rt},$$

where r is the constant risk-free interest rate.

This asset plays a central role in arbitrage pricing arguments.

1.2.3 Commodities

Commodities are physical goods such as oil, gold, or agricultural products. They are actively traded and their prices fluctuate due to supply and demand, geopolitical events, and macroeconomic conditions.

1.2.4 Currencies

Foreign exchange markets determine the price of one currency relative to another. Exchange rates fluctuate randomly and are often modeled similarly to stock prices.

2 Derivatives

A derivative is a financial contract whose value depends on (or is derived from) the value of another asset, called the *underlying*. If $S(t)$ denotes the price of the underlying asset, a derivative contract typically specifies a payoff at time T of the form,

$$\text{Payoff} = f(S(T)),$$

where f is a given function. In this course, the derivative payoff will be treated as a random variable, and our objective is to determine its fair price at time $t < T$.

2.1 Forward Contracts

A forward contract is an agreement entered at time 0 to buy (or sell) an asset at a predetermined future time T for a fixed price K . The payoff at maturity is,

$$S(T) - K$$

for the long position. Forward contracts are typically used for hedging price risk. For example, a firm expecting to buy a commodity in the future may lock in its price today to eliminate uncertainty.

2.2 Options

An option gives its holder the *right*, but not the obligation, to transact at a predetermined price.

European Call Option:

A European call with strike K and maturity T has payoff

$$(S(T) - K)^+ = \max(S(T) - K, 0).$$

European Put Option:

A European put with strike K and maturity T has payoff

$$(K - S(T))^+.$$

Unlike forwards, options create nonlinear payoffs.

2.3 Why Do Derivatives Exist?

Derivatives serve several important economic purposes.

2.3.1 Hedging

Hedging refers to reducing or eliminating unwanted risk. For example, an exporter expecting to receive foreign currency in the future may use a forward contract to lock in an exchange rate today. An investor holding stock may buy a put option to insure against large losses.

2.4 Speculation

Derivatives allow investors to take leveraged or nonlinear positions on future price movements without purchasing the underlying asset directly.

3 Forward Pricing and the Principle of No-Arbitrage

We now introduce the central idea that will drive all pricing results in this course: the principle of no-arbitrage and the method of replication.

An arbitrage opportunity is a trading strategy that requires no initial investment, never produces a loss, and has a strictly positive payoff with positive probability. In competitive financial markets such opportunities cannot persist. If they did, investors would exploit them immediately, forcing prices to adjust. The assumption that arbitrage does not exist is the basic economic axiom underlying modern mathematical finance.

The key logical implication of no-arbitrage is the following: if two portfolios produce exactly the same payoff at maturity, then they must have the same price today. Otherwise one could buy the cheaper portfolio and sell the more expensive one, locking in a riskless profit. This idea is called replication. If we can construct a trading strategy that replicates the payoff of a derivative, then the cost of constructing that strategy must equal the derivative's price.

We now apply this reasoning to price a forward contract. Consider a forward contract initiated at time 0 with maturity T and delivery price K . The payoff of a long forward position at time T is

$$S(T) - K.$$

We construct a portfolio that produces the same payoff.

At time 0, buy one unit of the stock for $S(0)$ and finance this purchase by borrowing $S(0)$ in the money market at the risk-free rate r . The initial net investment is zero, since the stock purchase is exactly financed by borrowing. At time T , the stock is worth $S(T)$, while the loan must be repaid in the amount $S(0)e^{rT}$. Therefore the terminal value of this strategy is

$$S(T) - S(0)e^{rT}.$$

Comparing this with the forward payoff, we see that if the delivery price satisfies

$$K = S(0)e^{rT},$$

then the forward payoff coincides exactly with the payoff of the replicating strategy. By the principle of no-arbitrage, the forward delivery price must therefore be

$$F_0 = S(0)e^{rT}.$$

It is important to notice what does *not* appear in this formula. The expected return μ of the stock plays no role. The forward price depends only on the current stock price and the risk-free rate. This illustrates a fundamental principle: derivative prices are determined by arbitrage considerations, not by expected returns.

To see why this must be true, suppose instead that the forward price were larger than $S(0)e^{rT}$. Then an investor could borrow $S(0)$, purchase the stock, and simultaneously enter a short forward

contract. At maturity, the stock would be delivered into the forward contract in exchange for the forward price, the loan would be repaid, and the difference would be a guaranteed profit. Similarly, if the forward price were smaller than $S(0)e^{rT}$, one could short the stock, lend the proceeds, and enter a long forward contract, again generating a riskless profit. Thus any deviation from $S(0)e^{rT}$ would create arbitrage.

We can also compute the value of a forward contract at an intermediate time $t < T$. Suppose the contract was initiated at time 0 with delivery price K . At time t , the stock price is $S(t)$, and the present value of paying K at time T is $Ke^{-r(T-t)}$. Using the same replication logic, the value of a long forward position at time t must be

$$f(t) = S(t) - Ke^{-r(T-t)}.$$

Again, this follows directly from constructing a portfolio consisting of one unit of the stock and borrowing the present value of K .

The forward contract is easy to price because its payoff is linear in the underlying asset and can be replicated by a simple buy-and-hold strategy. More complicated derivatives, such as options, have nonlinear payoffs and require dynamic rebalancing to replicate. The Black-Scholes-Merton equation that we derive next is nothing more than a continuous-time extension of this same replication principle.

4 Black-Scholes-Merton Equation

In the previous section we priced a forward contract by constructing a portfolio that replicated its payoff. The key idea was that if a derivative payoff can be replicated, then by the principle of no-arbitrage its price must equal the cost of the replicating strategy.

For a forward contract, replication was static: we bought the stock and held it until maturity. For options, however, the payoff is nonlinear in $S(T)$. A simple buy-and-hold strategy will not suffice. Instead, we must allow the portfolio to be rebalanced continuously over time. This leads to the idea of *dynamic replication*.

Assume that the stock price follows a geometric Brownian motion

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t), \quad S(0) > 0, \sigma > 0, \quad (1)$$

and that the money market account grows deterministically at rate r ,

$$B(t) = e^{rt}.$$

Evolution of a Self-Financing Portfolio

Let $X(t)$ denote the value of a portfolio that holds $\Delta(t)$ shares of the stock at time t , with the remaining wealth invested in the money market account. The process $\Delta(t)$ is assumed to be adapted, so that it depends only on currently available information.

The amount invested in the money market is

$$X(t) - \Delta(t)S(t).$$

Over an infinitesimal interval $[t, t + dt]$, the portfolio changes due to:

- capital gains on the stock position, $\Delta(t) dS(t)$,
- interest earned on the money market position, $r(X(t) - \Delta(t)S(t)) dt$.

Thus the portfolio satisfies the self-financing condition

$$\begin{aligned} dX(t) &= \Delta(t) dS(t) + r(X(t) - \Delta(t)S(t)) dt \\ &= \Delta(t)(\mu S(t) dt + \sigma S(t) dW(t)) + rX(t) dt - r\Delta(t)S(t) dt \\ &= rX(t) dt + \Delta(t)(\mu - r)S(t) dt + \Delta(t)\sigma S(t) dW(t). \end{aligned} \quad (2)$$

Discounted Quantities

As in forward pricing, it is convenient to discount by the money market account:

$$\tilde{S}(t) = e^{-rt}S(t), \quad \tilde{X}(t) = e^{-rt}X(t).$$

Applying Itô's formula to $e^{-rt}S(t)$ gives

$$\begin{aligned} d\tilde{S}(t) &= -re^{-rt}S(t) dt + e^{-rt}dS(t) \\ &= (\mu - r)e^{-rt}S(t) dt + \sigma e^{-rt}S(t) dW(t). \end{aligned} \quad (3)$$

Similarly,

$$\begin{aligned} d\tilde{X}(t) &= -re^{-rt}X(t) dt + e^{-rt}dX(t) \\ &= \Delta(t)(\mu - r)e^{-rt}S(t) dt + \Delta(t)\sigma e^{-rt}S(t) dW(t). \end{aligned} \quad (4)$$

Comparing (3) and (4), we obtain the compact relation

$$d\tilde{X}(t) = \Delta(t) d\tilde{S}(t). \quad (5)$$

Thus, after discounting, the portfolio evolves purely through exposure to the discounted stock. This is the continuous-time analogue of the replication logic used in forward pricing.

Evolution of the Option Price

Consider a European call option with payoff

$$(S(T) - K)^+.$$

We assume its price at time t can be written as

$$c(t, S(t)),$$

for a sufficiently smooth function $c(t, x)$.

Applying Itô's formula to $c(t, S(t))$ gives

$$dc(t, S(t)) = \left[c_t + \mu S c_x + \frac{1}{2} \sigma^2 S^2 c_{xx} \right] dt + \sigma S c_x dW(t). \quad (6)$$

Discounting again,

$$\begin{aligned} d(e^{-rt}c(t, S(t))) &= e^{-rt} \left[-rc + c_t + \mu S c_x + \frac{1}{2} \sigma^2 S^2 c_{xx} \right] dt \\ &\quad + e^{-rt} \sigma S c_x dW(t). \end{aligned} \quad (7)$$

Dynamic Replication

We now impose the replication requirement:

$$\tilde{X}(t) = e^{-rt}c(t, S(t)).$$

Therefore,

$$d\tilde{X}(t) = d(e^{-rt}c(t, S(t))).$$

Substituting (4) and (7) yields

$$\begin{aligned} \Delta(t)(\mu - r)S(t)dt + \Delta(t)\sigma S(t)dW(t) &= \left[-rc + c_t + \mu Sc_x + \frac{1}{2}\sigma^2 S^2 c_{xx} \right] dt \\ &\quad + \sigma Sc_x dW(t). \end{aligned} \tag{8}$$

Equating the stochastic terms gives

$$\Delta(t) = c_x(t, S(t)).$$

This choice eliminates the randomness of the hedged position, making it locally riskless. Substituting this into the drift terms gives

$$(\mu - r)Sc_x = -rc + c_t + \mu Sc_x + \frac{1}{2}\sigma^2 S^2 c_{xx}.$$

The terms involving μ cancel, leaving

$$\boxed{c_t + rSc_x + \frac{1}{2}\sigma^2 S^2 c_{xx} = rc.}$$

As in forward pricing, the expected return μ disappears. The derivative price is determined entirely by arbitrage considerations.

Terminal Condition

The PDE is solved subject to,

- Terminal condition:

$$c(T, x) = (x - K)^+.$$

- Boundary condition at $x = 0$:

$$c(t, 0) = 0,$$

since the option is worthless if the stock price is zero.

- Growth condition as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} [c(t, x) - (x - e^{-r(T-t)}K)] = 0,$$

reflecting that a deep in-the-money call behaves like a forward contract.

The unique solution to the PDE with the above conditions is

$$c(t, x) = xN(d_+(T-t, x)) - Ke^{-r(T-t)}N(d_-(T-t, x)), \tag{9}$$

where

$$d_{\pm}(\tau, x) = \frac{1}{\sigma\sqrt{\tau}} \left(\log \frac{x}{K} + (r \pm \frac{1}{2}\sigma^2)\tau \right),$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function.

5 Put-Call Parity

We now establish a fundamental relationship between European call and put options known as *put-call parity*. This identity follows directly from the principle of no-arbitrage and the replication arguments developed earlier.

Value of a Forward Contract

Recall that a forward contract with delivery price K obligates its holder to buy one share of stock at time T in exchange for payment K . The payoff at maturity is

$$S(T) - K.$$

Let $f(t, x)$ denote the value at time t of a forward contract when the stock price is $S(t) = x$. Using the replication argument from the previous section, the value of the forward contract is

$$f(t, x) = x - e^{-r(T-t)}K.$$

In particular, the *forward price* at time t is the value of K that makes the forward contract have zero value at time t . Solving

$$x - e^{-r(T-t)}K = 0$$

gives

$$F(t) = e^{r(T-t)}x.$$

Note carefully that the forward price is not the price of a forward contract. The forward contract has value zero when initiated with delivery price $F(t)$.

A Key Algebraic Identity

For any real number x ,

$$x - K = (x - K)^+ - (K - x)^+.$$

Indeed,

- If $x \geq K$, then $(x - K)^+ = x - K$ and $(K - x)^+ = 0$.
- If $x \leq K$, then $(x - K)^+ = 0$ and $(K - x)^+ = K - x$.

In either case, the identity holds.

Payoff Comparison at Maturity

Let

- $c(t, x)$ denote the value of a European call with strike K and maturity T ,
- $p(t, x)$ denote the value of a European put with strike K and maturity T .

At maturity T , we have

$$\begin{aligned}c(T, S(T)) &= (S(T) - K)^+, \\p(T, S(T)) &= (K - S(T))^+.\end{aligned}$$

Using the algebraic identity above,

$$S(T) - K = (S(T) - K)^+ - (K - S(T))^+.$$

Hence,

$$S(T) - K = c(T, S(T)) - p(T, S(T)).$$

But $S(T) - K$ is precisely the payoff of a forward contract with delivery price K . Therefore, at maturity,

$$f(T, S(T)) = c(T, S(T)) - p(T, S(T)).$$

No-Arbitrage Argument

Since the forward contract and the portfolio consisting of

- long one call,
- short one put,

have identical payoffs at time T , the principle of no-arbitrage implies that their values must agree at all earlier times:

$$f(t, x) = c(t, x) - p(t, x), \quad 0 \leq t \leq T.$$

Substituting the expression for the forward value,

$$x - e^{-r(T-t)}K = c(t, x) - p(t, x).$$

Rearranging, we obtain the *put-call parity formula*:

$$\boxed{c(t, x) - p(t, x) = x - Ke^{-r(T-t)}}.$$

Equivalently,

$$\boxed{c(t, x) + Ke^{-r(T-t)} = p(t, x) + x}.$$

Interpretation

Put-call parity states that:

- A long call and short put position replicates a forward contract.
- A call plus a bond equals a put plus the stock.

This relationship depends only on no-arbitrage and the ability to borrow and lend at the risk-free rate. It does not depend on any assumption about volatility or the specific dynamics of the stock price.

Exercises

1. Suppose $dS_t = \mu(t, S_t)dt + \sigma(t, S_t)dW_t$, where $\mu(t, S_t)$ and $\sigma(t, S_t)$ are sufficiently smooth functions of time and asset price. There is a continuous compounding risk-free rate of r available in the market. Derive the pricing PDE of a European call option with the terminal conditions.
2. Suppose the risk-free rate, $r(t)$, is a deterministic function of time. Assume that the stock price follows geometric Brownian motion. Derive the pricing PDE for a European call option.
3. Prove that two assets driven by a geometric Brownian motion with $\mu_1 \neq \mu_2$, but have identical volatility σ , will have identical European option price if the present values of the asset at the time are equal.
4. (*Constant elasticity of variance model*) Let,

$$dS_t = \mu S_t dt + \sigma S_t^\gamma dW_t$$

where $\gamma \geq 0$, then S_t is said to follow the CEV process. Derive the pricing PDE for a European call option on an asset driven by the CEV process.

5. In BSM, what happens to the call price as $T \rightarrow \infty$?
6. What happens to the call price in BSM as i) the stock price increases, ii) σ increases, iii) r increases, iv) K increases?